

NGR Operating Architecture Proposal

Protected Manual and Protected Automatic Modes

Proposal based on the May 16, 2026 design discussion

1. Core Proposal

The NGR locomotive control system should eventually support four operating modes selected from the individual locomotive's Blynk dashboard. These modes preserve the clear separation between direct engineer control and automatic railway operations, while adding a useful middle ground: manual or simple automatic running that remains visible to, and protected by, the railroad's block and dispatcher logic.

The central idea is that an operator should be able to interact with automatic train operations without becoming an untracked 'cowboy' on the railroad. A locomotive engaged in switching or local work can remain operator-driven, while the rest of the system still knows where it is and can protect both that locomotive and through traffic.

2. Four-Mode Operating Model

Mode	Motion authority	Block-aware	Auto restart after hold
Manual	Human	No	No
Protected Manual	Human	Yes	No
Protected Automatic	Local automatic cruise + Dispatcher	Yes	Yes
Auto	CTO / CE system	Yes	Yes, under CTO / CE logic

This table captures the proposed division of authority. The important distinction is not simply whether a locomotive is moving automatically. The distinction is who owns motion authority, whether the locomotive participates in railroad safety awareness, and who is allowed to restart it after a protective hold.

3. Mode Definitions

3.1 Manual

Manual mode remains the current direct locomotive-driving experience. The operator uses the locomotive dashboard for throttle, direction, brake, and emergency stop. The locomotive does not participate in automatic block logic, does not need to process station magnets for railroad coordination, and is not represented as an automatic traffic participant.

- Best for pure hands-on running.
- Appropriate when the operator is visually supervising the railroad directly.
- No automatic block protection is implied.

3.2 Auto

Auto mode is the existing dispatcher-controlled automatic operating world: CTO, Circuit Express, and later automatic service profiles. Once Auto is selected on the locomotive dashboard, the operator leaves that dashboard for normal operation and controls the trains from the Dispatcher dashboard.

- CTO / CE logic owns train motion.
- Dispatcher commands govern automatic running.
- The locomotive remains fully involved in peer communication, station magnet logic, and safety logic appropriate to automatic service.

3.3 Protected Manual

Protected Manual is a proposed third mode. The operator remains on the same locomotive dashboard used for ordinary manual operation and still controls throttle, direction, brake, and emergency stop. However, the locomotive also participates in the railway's safety logic.

- The locomotive reads magnets and establishes block position.
- It broadcasts its position so other locomotives can see it in their peer tables.
- It respects dispatcher-closed blocks and occupied-block prohibitions.
- If the operator commands a move that would enter a prohibited block, the locomotive remains stopped and reports the reason locally on its dashboard.
- If it is held for protection, it does not restart itself. The operator resumes motion deliberately.

This mode is especially suitable for local switching work. It allows a human engineer to perform shunting and local movements while automatic trains continue to run elsewhere under protection.

3.4 Protected Automatic

Protected Automatic uses the same block awareness and dispatcher protection as Protected Manual, but changes restart authority. The operator prepares the locomotive locally - for example, sets forward direction and a desired throttle or cruise level - and then the locomotive may proceed automatically within the protection rules.

- The locomotive remains block-aware and peer-visible.
- Dispatcher or occupancy logic may stop it when the path ahead is not clear.
- When the protective hold clears, the locomotive may restart automatically.
- It is not CTO and it is not Circuit Express; it is a simpler autonomous road-running mode under dispatcher protection.

This mode would be useful for a locomotive that should circulate as a separate train without participating in the CTO bubble choreography.

4. Dashboard and Operator Experience

Mode selection should live on each locomotive's own Blynk dashboard. This preserves the intuitive relationship between the operator and the locomotive. The dashboard remains the operating surface for Manual, Protected Manual, and likely the setup surface for Protected Automatic. Only full Auto mode hands operational authority over to the Dispatcher dashboard.

Mode selected on locomotive dashboard	Operator remains on locomotive dashboard?	Primary operating surface
Manual	Yes	Locomotive dashboard
Protected Manual	Yes	Locomotive dashboard
Protected Automatic	Mostly setup there; execution is locally automatic	Locomotive dashboard + Dispatcher protections
Auto	No, after handoff	Dispatcher dashboard

The operator experience should match the railroad concept. Protected Manual should feel like: 'I am still the engineer of this locomotive, but I am operating under railroad authority.' Auto should feel like: 'This locomotive is now part of the dispatcher-controlled automatic railroad.'

5. Dispatcher Block Closure / Local Work Authority

Protected Manual naturally leads to a dispatcher 'work zone' or block-closure function. This is closely related to the previously discussed torpedo / reserved-block idea, now expressed in a more general operating architecture.

- The Dispatcher dashboard offers a request to close a named block for local work.
- Automatic trains approaching that block are stopped or withheld before entry.
- A Protected Manual locomotive may perform switching within the reserved area.
- The local locomotive remains visible through block broadcasts and peer tables.
- Once the protected locomotive leaves the closed block, the Dispatcher may automatically clear the closure and release waiting traffic, if that behavior is later chosen.

In this mode the Dispatcher is genuinely acting as a dispatcher: it does not drive the local locomotive, but it governs track authority and protects interactions between local work and through traffic.

6. Relationship to CTO Bubbles

This proposal avoids forcing a third locomotive into the internal CTO leader/trailer choreography. The CTO bubble remains a two-locomotive operating unit with its present elegant choreography. Other trains may coexist with that bubble through block awareness and dispatcher authority rather than through a more complicated three-locomotive CTO chain.

- CTO bubble = internal paired behavior between two locomotives.
- Protected Manual / Protected Automatic locomotive = separate traffic participant, visible to the system.

- Dispatcher protection = the layer that prevents the separate participant and the bubble from violating occupancy or work-zone rules.

This matches the emerging NGR design principle: keep the specialized automatic behaviors elegant, and use a higher-level safety layer to manage interactions between operating units.

7. Safety Logic Proposed

The safety objective is not to remove human operation, but to prevent a manually controlled locomotive from becoming invisible to the automatic railroad. A protected mode locomotive should therefore satisfy two conditions:

- It must know its own block position after magnet detection.
- It must make that position visible to the rest of the system.

From that foundation, block-entry restrictions can be added. Examples include:

- Do not enter a dispatcher-closed block.
- Do not proceed into a block that is already occupied by an active peer, depending on the exact occupancy rule adopted.
- If movement is denied, hold motor target at zero and display a local warning such as BLOCK AHEAD OCCUPIED or BLOCK CLOSED FOR LOCAL WORK.

Protected Manual and Protected Automatic can share most of this safety logic. Their crucial difference is restart authority:

- Protected Manual: the operator must resume motion after a hold.
- Protected Automatic: the system may resume motion when the route becomes clear.

8. Why This Fits the NGR Philosophy

This proposal is compelling because it models railroad operations after the real thing. A local operator can work a train, a dispatcher can grant or restrict authority, and through traffic can continue under automatic protection. The operator participates in the logic of the railroad rather than operating outside it.

- It preserves satisfying hands-on locomotive control.
- It preserves the beauty of existing CTO and CE automatic operations.
- It provides a disciplined way for manual work to coexist with automatic running.
- It creates a future path for Oscar, Gunther with a sensing car, or other local locomotives to join operating sessions in a more realistic way.

9. Suggested Development Sequence

1. Record this as a future architecture decision, not an immediate disruption to the now-working CTO / CE platform.
2. Define the mode-state model on the locomotive dashboard: Manual, Protected Manual, Protected Automatic, Auto.
3. Specify exactly what magnet processing and peer-table broadcasting are active in each protected mode.
4. Define the first minimal block-protection rule for Protected Manual.
5. Define the Dispatcher block-closure request and block-release behavior.
6. Only after the safety model is clear, prototype Protected Manual before Protected Automatic.
7. Treat Protected Automatic as the second step, using the same protection layer plus automatic restart authority.

10. Open Questions for Later Design

- What exact block-occupancy rule should prohibit a Protected Manual locomotive from moving?
- Should a block closure release automatically when the local locomotive reports a new block, or require dispatcher confirmation?
- How should a protected locomotive be staged when first entering the protected mode if it has not yet crossed a known magnet pair?
- Should Protected Automatic use a simple user-set cruise target, or a compiled profile?
- Should the dispatcher show a concise indicator that a locomotive is in Protected Manual or Protected Automatic?

11. Working Conclusion

The proposed four-mode architecture appears to be a natural extension of the current NGR control system. It does not replace the newly successful integrated Manual / Auto controller. Instead, it extends that integration by creating two intermediate modes that support realistic local operations under railroad authority.

The most important new idea is Protected Manual: a locomotive that remains engineer-driven from its own dashboard, but is no longer invisible to the automatic railroad. That mode would allow a human operator to interact with CTO and CE traffic in a way that is operationally rich, realistic, and safer than visually supervised free-running alone.